Reinforcement Learning for Multi-Agent Shepherding

Training autonomous agents to coordinate and herd in complex environments

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MOTIVATION & PROBLEM STATEMENT

Multi-Agent Control

Shepherding is a complex control problem requiring precise coordination, spatial awareness, and long-horizon planning to influence collective behavior.

Practical Applications

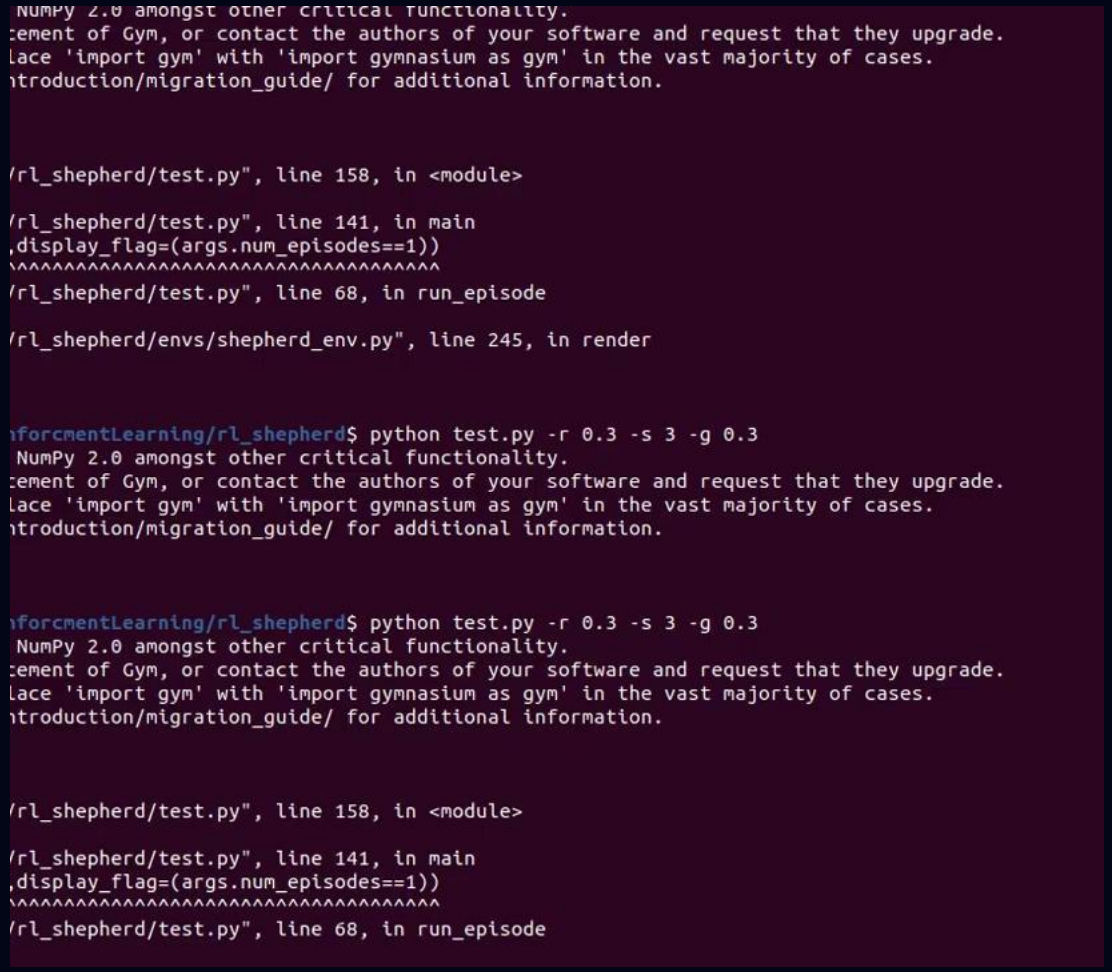
- Autonomous Logistics & Traffic Control
- Collective Agent Interaction Modeling
- Safety-critical crowd management

ENVIRONMENT OVERVIEW

2D Continuous Dynamics

A physical simulation where agents interact through proximity forces.

- **Entities:** Shepherd agents, 1–3 Sheep, and a specific Goal region.
- **Reactive Physics:** Sheep respond to shepherd proximity via repulsive forces.
- **Action Space:** Continuous force vectors controlling shepherd movement.



PROGRESSIVE CURRICULUM DESIGN

L1: Basic

Deterministic. Sleepy sheep. Learn fundamental interaction.

L2: Active

Stochastic. Sheep move randomly. Learn robustness to noise.

L3: Obstacles

Spatial constraints. Circular obstacles block sheep. Learn navigation.

L4: Multi-Agent

Alternating Control. Two shepherds act turn-based. Learn coordination and teamwork.

BASELINE COMPARISON MODELS

Lazy Agent

Constant action output. Serves as a baseline for minimal environment noise and drift.

Tipsy Agent

Random movement. Measures environment robustness and baseline stochastic success.

Rule-Based

Targets furthest sheep from goal. Excellent in simple env; suboptimal in complex/competitive env.

REINFORCEMENT LEARNING APPROACH

PPO (Policy Gradient)

On-policy learning with clipped surrogate objectives. Ideal for continuous control and stability in the movement force space.

DQN (Value-Based)

Off-policy Bellman updates. Used with discretized action spaces to compare learning efficiency in lower-dimensional tasks.

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Observation: [Shepherd_Sheep, Shepherd_Sheep, Goal_Shepherd, Obstacle_Shepherd]
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REWARD DESIGN

- **Shaping Rewards:**

  +300.0 × Progress (Mean distance reduction).

  +5.0 × exp(-5 × dist_to_target_sheep).

- **Penalties:**

  -10.0 × exp(-100 × movement) [Regularization].

  -0.02 Step penalty | -10.0 Time Limit penalty.

- **Termination:**

  +100.0 Per sheep reached goal.

  +200.0×N + 5.0×(Remaining Steps) [Completion Bonus].

MATHEMATICAL REPRESENTATION

$$R_{\text{total}} = 300\Delta\bar{d} + 5e^{-5d_{\text{target}}} - 10e^{-100\|v\|} - 0.02$$

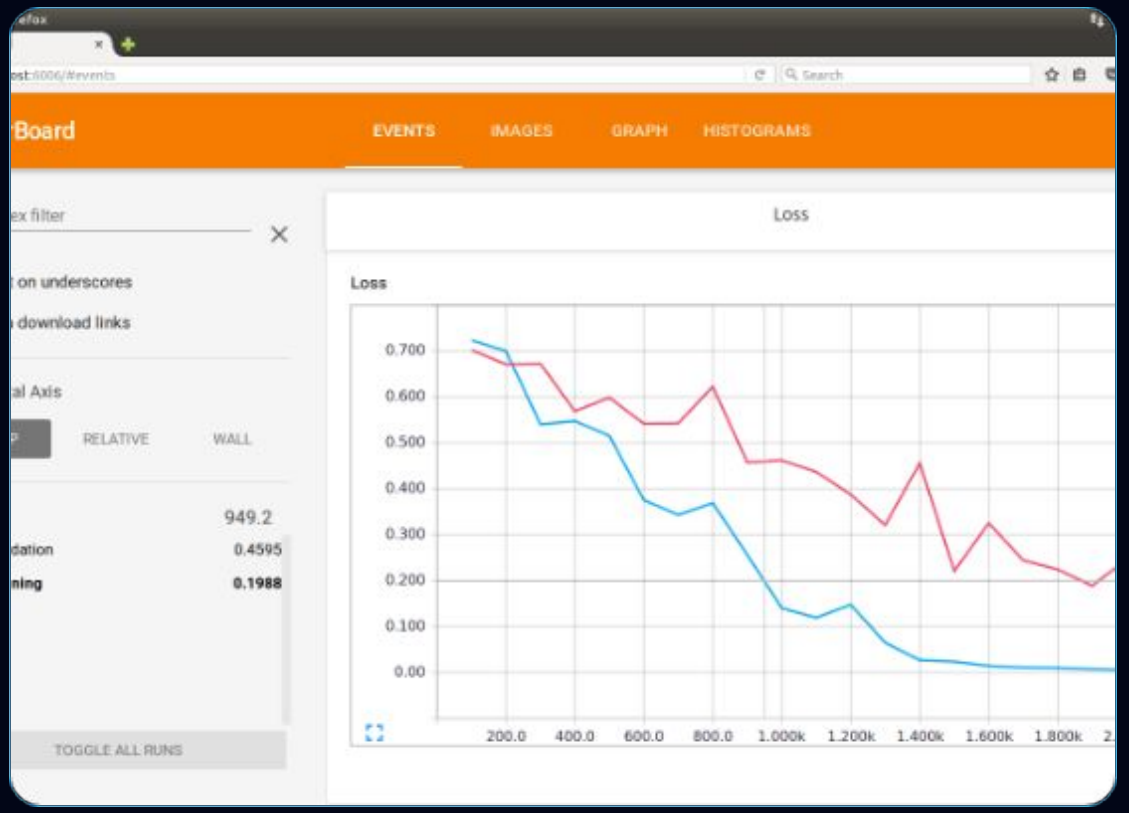
</> target_sheep is the furthest sheep from the goal. This encourages the agent to focus on the "worst case" entity.

TRAINING MONITORING: TENSORBOARD

Real-time Diagnostics

TensorBoard is utilized to track convergence and catch instability early.

- **Scalar Tracking:** Episode reward, policy loss, and value loss.
- **Hyperparameters:** Comparing learning rates and entropy coefficients across runs.
- **Debugging:** Monitoring policy entropy to prevent premature convergence or "agent collapse."



EVALUATION BEST PRACTICES

"Do NOT rely on average reward alone. Verify performance by running 50+ episodes per level with fixed random seeds, comparing explicitly against Lazy, Tipsy, and Rule-Based baselines."

SYSTEM LIMITATIONS

Scalability

State space complexity explodes with many sheep (>5).

Reward Tuning

Highly sensitive to reward shaping coefficients.

Compute Cost

Training multi-agent levels requires high GPU iterations.

Conclusion & Questions

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